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TRANSFORMER X/R RATIO CALCULATION

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TRANSMISSION ENGINEERING STANDARD

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1.0 SCOPE

The purpose of this standard is to define the criteria, effects and calculation for *X/R* ratio in an electrical system with particular reference to power transformers of all voltage levels to be used in transmission network of Saudi Electricity Company (SEC), Kingdom of Saudi Arabia.

2.0 INTRODUCTION

In an electrical system, impedance has two components. The first one is called reactance *(X)*. Reactance depends upon inductance and frequency. Inductance reflects how hard it is to change the current. All conductors have some inductance, but more useful example of a component having inductance is a coil of wire as used in transformers, reactors, motors and generators etc. Power frequency is fixed at either 60 or 50 Hz, depending upon where in the world the electrical system is, so the reactance is solely dependent upon the inductance.

The second component of impedance is the familiar resistance *(R)*. Resistance is a measure of how hard it is for the current to flow. When the current flows through a material having resistance, heat is transferred from the material to the surroundings.

X/R ratio is simply the ratio of a system reactance to the system resistance, looking back towards the power source from any point in a power circuit, assuming that a short circuit is applied to the system at that point. It is another way of stating the power factor of the source system. Mathematically, as power factor is equal to the $\cos\theta°$, the *X/R* is equal to $\tan\theta°$, i.e

$$\theta° = \tan^{-1} X/R$$
$$X = Z \,.\, \sin\theta°$$
$$R = Z \,.\, \cos\theta°$$
$$p.f = \cos\theta° \;\; or \;\; \cos[\tan^{-1}(X/R)]$$

This is the power factor of the system to that point. It has absolutely no relationship to the power factor of any load on the system. Since the generators, transformers and the transmission lines are quite highly inductive, the *X/R* ratio is generally significantly above unity in any utility or industrial power system.

3.0 ROLE OF *X/R* RATIO

When a short circuit occurs, the rms value of the symmetrical fault current is determined by the source system voltage and the total system impedance to the point of fault. However, all faults involve significant asymmetry in at least one phase. The asymmetry is treated in the analysis as a dc component which is combined with the ac symmetrical component to give a new current value, the rms asymmetrical value. The largest asymmetrical fault current occurs when a fault happens at a point when the voltage is zero. Then, the asymmetrical fault current depends only on the *X/R* ratio, or power factor, and the magnitude of the symmetrical fault current.

The dc component of a fault current decays rather rapidly, reaching an insignificant value in a matter of three to five cycles of the power frequency. However, this rate of decay is determined by the *X/R* ratio of the circuit at the point of fault. The higher the *X/R* ratio, the more slowly the dc component decays.

4.0 ASYMMETRY FACTOR & PEAK FACTOR

The factor k or asymmetry factor accounts for the initial offset of the test current and $\sqrt{2}$ accounts for the peak-to-rms value of a sinusoidal wave. The factor $k \times \sqrt{2}$ or peak factor K, depends on the ratio X/R and is based upon the following expression:

$$K = k \times \sqrt{2} = \{ 1+[e^{-(\varphi+ \pi/2)R/X}]\sin\varphi\} \times \sqrt{2}$$

Where:

k asymmetry factor
e is base of the natural logarithm;
φ is the phase angle which is equal to arctan X/R, in radians.

The amplitude I_{asym} of the first peak of the asymmetrical test current is calculated as follows:

$$I_{asym} = I_{sym} \times k \times \sqrt{2}$$

or

$$I_{asym} = I_{sym} \times K$$

where I_{sym} is the rms symmetrical short circuit current.

For the details of calculation for asymmetrical short circuit current refer TES-P-119.37. (Short Circuit Calculation for Substation main equipments). A Table for the peak factor K for the different values of X/R ratios has also been given in TES-P-119.37. Suitable value of K can be selected from this Table.

The ability of the power transformers to withstand the dynamic effects of the short circuit current is demonstrated either by tests or by calculations. For two winding and three winding transformers the short circuit current and asymmetrical short circuit current shall be calculated per formula given in TES-P-119.37.

5.0 RANGE OF TRANSFORMER *X/R* RATIOS

5.1 On a power system, the *X/R* ratios are the highest for the generating units and the transformers and lower for the transmission lines. An investigation for the typical range of *X/R* ratios carried out by a power engineer of a leading manufacturing company of substation electrical equipment, for over one hundreds power transformers including all sizes from 12 MVA to 800 MVA, core form and shell form, all voltage range, generator step-up units (GSU) vs Autos, range of load loss evaluation, etc. reveals the following result:

		Range of *X/R* Ratio
Small power transformers	(25 kVA to 2500 kVA)	15 to 40
Medium power transformers	(2501 kVA to 100 000 kVA)	35 to 70
Large power transformers	(Above 100 000 kVA)	40 to 125

This comparison shows that the *X/R* ratio for the power transformers can be anywhere between 15 and 125, lower for smaller transformers and higher for the larger transformers.

5.2 Typical values of *X/R* ratios of power transformers can be obtained from the Figure 40-1. Based on the class of transformer, proper factor shall be selected from the Table 40-1 below. Transformer MVA rating shall be multiplied by this factor before using Figure 40-1 to obtain the typical value of *X/R* ratio.

Table 40-1

Class	Rating in MVA	Factor
ONAN	All ratings	1.67
ONAF	Up to 14.9	1.33
ONAF	16 & up	1.25
OFAF	All ratings	1.00

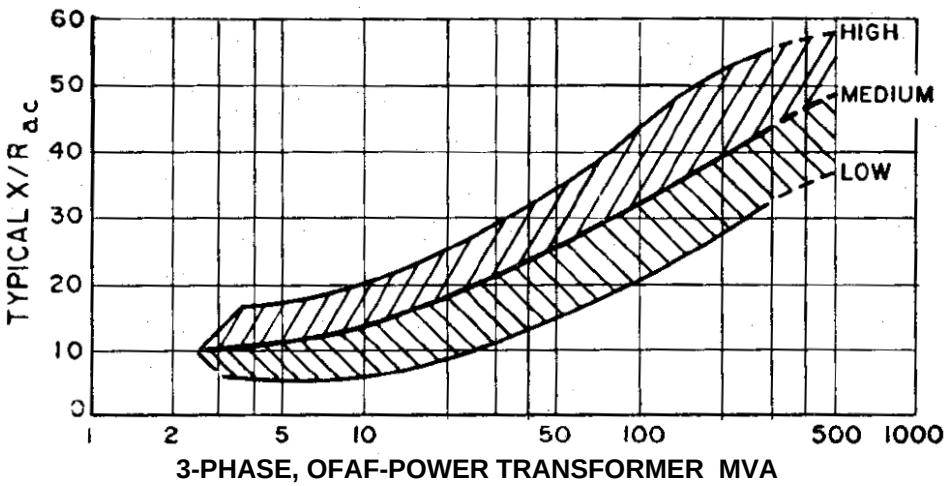

Figure 40-1 : X/R Range for Power Transformers at 60 Hz

5.3 When losses are not given in the nameplate then Power Transformer *X/R* ratio shall be calculated as follows:

a. Transformer winding resistance measurement at site.

b. Referring the measured resistance value to primary or secondary side of Power Transformer.

c. Converting the measured resistance to 75°C reference value.

d. Converting the resistance value at 75°C to percentage value by the following formula:

$$\% R = \frac{R}{kV^2 / MVA} x100$$

Where

R = Transformer Winding Resistance in Ohms

Percentage Reactance is given by the formula

$$\% X = \sqrt{Z^2 - R^2}$$

Alternatively, ignoring stray losses, the square of rated current times phase winding resistance of each respective winding when added together would yield copper losses which can be then used for calculation of Power Transformer X/R as shown in the following typical examples in clause 5.4

5.4 Calculation of *X/R* Ratio of Power Transformers in SEC System (Typical)

5.4.1 380/132 kV, 500 MVA Power Transformer

Data specified by SEC

i. Impedance of HV-LV at principal tap 22 % at 500 MVA base

ii. Transformer Rating 300/400/500 MVA

Data by Transformer Manufacturer

iii. Guaranteed Load Losses (500 MVA Base) 1250 kW
(Typical value)

iv. *X/R* Calculation:

Resistance *'R'* = (1250 kW/500,000kVA)×100=0.25 %

Value of X $= \sqrt{Z^2 - R^2}$

Where Z is percentage impedance of transformer

Hence X $= \sqrt{22^2 - 0.25^2}$

≈ 22

Reactance *'X'* = 22 %

X/R ratio = 22/0.25

= 88

5.4.2 132/13.8 kV, 50/67 MVA Power Transformer

Data specified by SEC

i. Impedance of HV-LV at principal tap
at 50 MVA base Z_{50} 22 %

ii. Impedance of HV-LV at principal tap at 67 MVA base Z_{67} = $(Z_{50}/50) \times 67$
= 29.48 %

iii. Transformer Rating 50/67 MVA

Data by Transformer Manufacturer

iv. Guaranteed Load Losses (67 MVA Base) 395 kW
(Typical value)

v. *X/R* Calculation:

% Resistance *'R'* =(395 kW/67,000 kVA)×100=0.589 %

(Value of resistance is very less compared to transformer percentage impedance and hence can be neglected for *X* calculation)

Reactance *'X_{67}'* = 29.48 %

X/R ratio = 29.48/0.589

= approx. 50

6.0 INTERPRETATION

The *X/R* ratio of the power transformers varies over a wide range. Generally, the *X/R* ratio is lower for small transformers and higher for larger transformers based on the MVA ratings. For a specified value of the transformer percentage impedance, losses shall be higher with the lower *X/R* ratio and vice versa.

The higher the *X/R* ratio, the higher the asymmetrical fault current. Accordingly, the equipment ratings shall be verified for their rated peak withstand current capability in addition to the rated short-time withstand current.

A higher *X/R* ratio, with its slower decay rate, will result in a higher asymmetrical fault current at contact parting time. If the *X/R* ratio is too high, the asymmetrical fault current may exceed the breaker's interrupting capacity. Accordingly, breaker's interrupting capacity shall be verified with reference to excessive *X/R* ratio.

Briefly, the transformer *X/R* ratio shall be selected in view of the impedance and full load losses of the transformer, short-time and peak withstand current ratings of the associated equipment and interrupting capability of the switching devices.

7.0 BIBLIOGRAPHY

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